

Machine Learning driven Emergency Department Length of Stay, Admission, and Hospitalization Prediction Using Triage-Only Data

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2 ABSTRACT

3 **Background:** Emergency Department (ED) crowding is a critical public health challenge.
4 Prolonged clinic stay (≥ 8 hours) affects over 20% of visits. Hospital admission and prolonged
5 hospitalization (> 7 days) consume disproportionate resources. Clinicians currently rely on triage
6 scales like the Emergency Severity Index (ESI), which have limited predictive accuracy. Machine
7 learning (ML) using triage-only data could support these decisions, but evidence on model
8 accuracy remains limited.

9 **Methods:** Using the MC-MED dataset (118,385 adult ED visits), we evaluated four ML models
10 (XGBoost, LightGBM, Random Forest, TabNet) on three binary classification tasks using 29
11 triage-available variables to predict: (1) prolonged ED length of stay (ED-LOS ≥ 8 hours vs. < 8
12 hours), (2) hospital admission (admit vs. discharge), and (3) prolonged hospital length of stay
13 (hospital LOS > 7 days vs. ≤ 7 days) among admitted patients. All 29 features were available
14 within 90 minutes of arrival. Area under the receiver operating characteristic curve (AUC-ROC)
15 was the primary performance metric. SHAP (SHapley Additive exPlanations) analysis identified
16 the most important features for each prediction task.

17 **Results:** For ED-LOS ≥ 8 hours, XGBoost achieved an AUC-ROC of 0.660 (95% CI: 0.652-
18 0.668). For hospital admission, XGBoost achieved an AUC-ROC of 0.817 (95% confidence
19 interval [CI]: 0.810-0.824), an improvement over the ESI baseline of 0.678. For hospital LOS
20 > 7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.671 (95% CI:
21 0.662-0.680). SHAP analysis identified consistent top predictors across tasks: triage acuity level,
22 emergency medical services (EMS) arrival, age, disease condition count, and the Charlson
23 Comorbidity Index.

24 **Conclusion:** ML using only triage-available data accurately predicts three critical ED binary
25 treatment outcomes. SHAP analysis provides interpretable feature importance to support clinical
26 decision-making.

27 **Keywords:** machine learning, emergency department, hospital admission, length of stay, Shapley
28 Additive Explanations

1 INTRODUCTION

29 Emergency departments (EDs) offer a fundamental service, providing emergency treatments 24 hours a
30 day, 365 days a year, therefore representing a critical public health service role [1, 2, 14]. Crowding
31 in EDs has been recognised as a critical challenge in hospital administration [3, 5]. ED crowding has
32 been linked to increased costs, longer waiting times, higher mortality rates, and adverse outcomes for
33 vulnerable populations [7, 9, 33].

Emergency department length of stay (ED-LOS) is a significant indicator of ED bottlenecks. ED-LOS is defined as the interval from ED arrival to ED departure [13, 14]. A prolonged ED-LOS is a consequence of crowding and has been associated with a 37% elevated mortality risk among admitted patients [7]. Several studies have defined an extended ED-LOS as more than 3 hours [14], while others use thresholds of 4 to 12 hours [19]. Prolonged clinic stay not only affects patient outcomes but also contributes to ambulance diversion, treatment delays for time-sensitive conditions, and inefficient resource utilization across the hospital system.

Clinicians currently rely on triage scales such as the Emergency Severity Index (ESI) to prioritize patients and predict clinical trajectories. However, the ESI demonstrates limited predictive accuracy for key outcomes such as hospital admission and prolonged length of stay when used in isolation, with AUC-ROC values typically below 0.70 [20]. This limitation creates an opportunity for data-driven approaches that integrate multiple patient characteristics available at triage.

Machine learning (ML) techniques are frequently used in healthcare for diagnosis, forecasting patient outcomes, and resource allocation [8, 17]. Following this approach, various studies have examined ways to predict ED-LOS, hospital admission, and hospital LOS using ML strategies [16, 18, 24]. Ricciardi et al. [14] achieved 74.8% accuracy for predicting ED-LOS greater than 3 hours using Random Forest. Wu et al. [16] found that LightGBM outperformed other models for continuous ED-LOS prediction in a Chinese hospital setting.

However, existing models have three critical limitations. First, most studies predict ED-LOS, hospital admission, or hospital LOS in isolation, despite their clinical interdependence in real-world emergency care pathways [13]. Second, many models incorporate laboratory results or specialist consultations that are unavailable at triage, limiting real-time clinical applicability [20]. Third, insufficient interpretability validation beyond single-model SHAP analysis raises concerns about algorithm-specific artifacts versus true clinical signals [26, 29].

Given these considerations, this work aims to build a predictive model to forecast three binary treatment outcomes using only triage-available variables: (1) prolonged ED-LOS (≥ 8 hours), (2) hospital admission, and (3) prolonged hospital LOS (> 7 days) among admitted patients.

Using the MC-MED dataset (118,385 adult ED visits from Stanford Health Care, 2020 to 2022), we evaluated four ML algorithms (XGBoost, LightGBM, Random Forest, and TabNet) on three binary classification tasks using 29 triage-available variables. Predicting these outcomes in advance would be beneficial for planning staff allocation and bed management with the goal of reducing ED crowding.

2 METHODS

2.1 Prediction Tasks

Three binary classification tasks were formulated using only triage available variables. Task 1 predicted prolonged emergency department length of stay defined as eight hours or more versus less than eight hours [7, 19]. Task 2 predicted hospital admission, where admission included inpatient, observation, and intensive care unit dispositions, versus discharge to home [10, 18]. Task 3 predicted prolonged hospital length of stay defined as more than seven days versus seven days or less, and this task was applied exclusively to admitted patients [13, 51].

2.2 Data Resource and Preprocessing

This study analysed the MC-MED dataset [10, 11], which contains 118,385 adult emergency department visits from Stanford Health Care between January 2020 and December 2022. The dataset includes demographics, triage vital signs, chief complaints, comorbidity measures, home medications, and visit outcomes [12].

Figure 1 shows the data preprocessing pipeline. The pipeline consisted of five sequential stages. Stage one involved raw data loading from the three source tables provided with the MC-MED dataset: `visits.csv`, `meds.csv`, and `pmh.csv`. These tables were merged using the Contact Serial Number to link visit specific records and the Medical Record Number to aggregate longitudinal medication and past medical history data. Stage two applied the chronological split into training, validation, and test sets

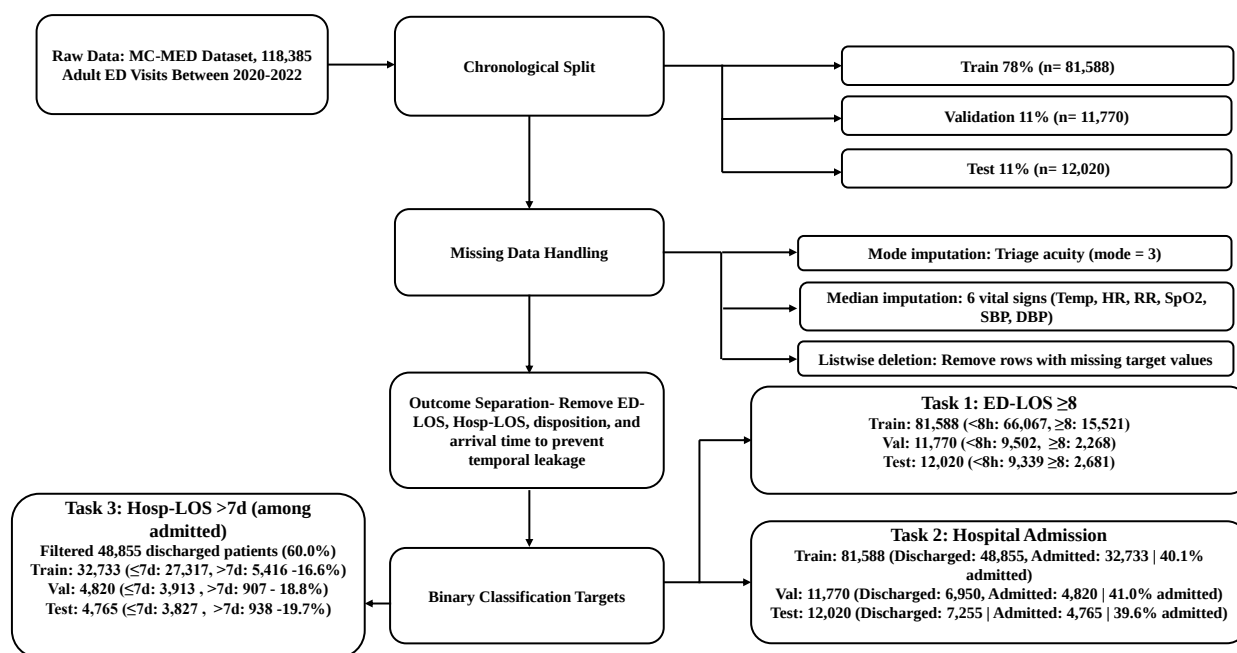


Figure 1. Chronological preprocessing pipeline for the MC-MED dataset showing split proportions, missing data handling methods (median, mode), outcome separation, and three binary classification targets.

82 using the pre defined split files provided with MC-MED, which ensured temporal separation between
 83 periods with no patient appearing in more than one split. Stage three handled missing data using the
 84 procedures described in detail below. Stage four performed outcome separation, where all time dependent
 85 columns including emergency department length of stay, hospital length of stay, disposition status, arrival
 86 time, admit time, discharge time, and departure time were removed from the feature set before model
 87 training to prevent temporal and data leakage. Stage five defined the three binary classification targets as
 88 described in the preceding subsection.

89 2.2.1 Missing Data Handling

90 Missing data were handled using three distinct strategies applied sequentially, with all imputation
 91 parameters derived exclusively from the training set to prevent information leakage from the validation or
 92 test sets.

93 The first strategy addressed missing values in the triage acuity feature. This feature was stored in the
 94 raw data as a string combining a numeric prefix and a description, with the specific formats being “1-
 95 Resuscitation”, “2-Emergent”, “3-Urgent”, “4-Semi-Urgent”, and “5-Non-Urgent”. The numeric prefix
 96 from one to five was extracted by taking the first character of the string. Missing values in this feature,
 97 which affected 8.4 percent of training samples, were imputed using the mode calculated from the training
 98 set only. The mode value was three, corresponding to the “3-Urgent” triage level. This mode value was
 99 then applied to impute missing triage acuity values in the validation and test sets.

100 The second strategy addressed missing values in the six continuous vital sign features: temperature,
 101 heart rate, respiratory rate, peripheral oxygen saturation, systolic blood pressure, and diastolic blood
 102 pressure. Missingness rates for these features ranged from 4.2 percent to 8.9 percent of training samples.
 103 These missing values were imputed using feature wise median values. The SimpleImputer class from
 104 the scikit learn library [15] was employed with the strategy parameter set to ‘median’. The imputer was
 105 fitted exclusively on the training set, calculating the median for each vital sign feature from the training
 106 data only. The fitted imputer was then used to transform the validation and test sets, ensuring that no
 107 information from future or held out data influenced the imputation values.

Table 1. Descriptive characteristics of the MC-MED cohort by chronological split

Characteristic	Full Cohort	Train (78%)	Val (11%)	Test (11%)
N	118,385	81,588	11,770	12,020
Age, mean (SD)	52.3 (19.8)	52.1 (19.7)	52.5 (19.9)	52.4 (19.8)
Female (%)	52.1	52.0	52.2	52.1
Race (% White)	62.3	62.2	62.4	62.3
EMS arrival (%)	31.7	31.6	31.9	31.7
Admission rate (%)	28.3	28.2	28.5	28.3
ED-LOS 8h or more (%)	22.3	22.1	22.5	22.3
Hosp-LOS more than 7d (%)*	19.7	19.6	19.9	19.7
Charlson Index, mean (SD)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)

*Among admitted patients only, full cohort n=33,516, test set n=4,765

The third strategy handled missing target values. For each prediction task, any sample with a missing target value was removed from the analysis using listwise deletion. For Task 3, which predicted prolonged hospital length of stay among admitted patients only, this resulted in the exclusion of 142 training samples representing 0.4 percent of admitted patients where hospital length of stay was not recorded.

After applying these three strategies, no missing values remained in the feature set. All 29 features were complete for all samples across the training, validation, and test splits. This complete data matrix enabled direct compatibility with all four machine learning algorithms, including TabNet which cannot process missing values natively.

The dataset was chronologically split using the pre defined split files provided with MC-MED, which ensure temporal separation between training, validation, and test periods [10]. A total of 13,007 patient encounters were excluded from the pure chronological ordering because their Contact Serial Number was not present in the chronological split files. For Task 3, which predicted prolonged hospital length of stay among admitted patients, the analysis was further restricted to admitted patients only, resulting in training set size of 32,733, validation set size of 4,820, and test set size of 4,765.

Table 1 presents the demographic and clinical characteristics of the study cohort across the three chronological splits. The mean age was 52.3 years with a standard deviation of 19.8 years, with female patients accounting for 52.1 percent of visits. The baseline admission rate was 28.3 percent in the full cohort. The prevalence of prolonged emergency department length of stay, defined as eight hours or more, was 22.3 percent. The prevalence of prolonged hospital length of stay, defined as more than seven days among admitted patients, was 19.7 percent.

2.3 Feature Set

A total of 29 triage available features were extracted for each emergency department visit by integrating the three source tables. The resulting predictors were organized into six clinical domains: demographics, arrival context, triage vitals and acuity, chief complaint, comorbidity burden, and medication burden. All variables are routinely captured within the first 90 minutes of emergency department arrival, ensuring real time applicability for early clinical decision support [20]. To prevent temporal and data leakage, no post triage or outcome dependent variables including emergency department length of stay, hospital length of stay, disposition status, or post admission timestamps were included in the predictor set. Table 2 provides the complete feature list with data nature and derivation method.

2.4 Machine Learning Algorithms

Four machine learning algorithms were implemented: Random Forest, XGBoost, LightGBM, and TabNet. These algorithms follow different theoretical approaches [14, 16]. Random Forest uses bootstrapping to train numerous decision trees concurrently and combines their outputs using bagging [14]. XGBoost is a gradient boosting algorithm with regularization to prevent overfitting [29]. LightGBM is a histogram based gradient boosting algorithm optimized for efficiency [16]. TabNet is a deep learning architecture with attention mechanisms specifically designed for tabular data [24].

Table 2. Triage available features with derivation methods, N=29

Feature Name	Data Nature or Derivation
Demographics (7)	
Age	Direct from electronic health record, range 18 to 120 years
Sex (Male)	Direct from electronic health record, binary
Race (White)	Direct from electronic health record, binary
Race (Black)	Direct from electronic health record, binary
Hispanic Ethnicity	Direct from electronic health record, binary
Medicare Insurance	Direct from electronic health record, binary
Medicaid Insurance	Direct from electronic health record, binary
Triage Vitals and Acuity (7)	
Triage Temperature	Direct from electronic health record, degrees Celsius
Triage Heart Rate	Direct from electronic health record, beats per minute
Triage Respiratory Rate	Direct from electronic health record, breaths per minute
Triage SpO2	Direct from electronic health record, percentage
Triage Systolic Blood Pressure	Direct from electronic health record, mmHg
Triage Diastolic Blood Pressure	Direct from electronic health record, mmHg
Triage Acuity Level	Derived by extracting numeric prefix from strings “1-Resuscitation” through “5-Non-Urgent”, range 1 to 5
Arrival Context (3)	
Arrival Hour	Derived from Arrival_time timestamp, range 0 to 23
Weekend Arrival	Derived from Arrival_time, binary indicator for Saturday or Sunday
EMS Arrival	Direct from electronic health record, binary indicator for Means_of_arrival equal to “EMS”
Chief Complaint (7)	
Complaint Count	Derived as number of presenting complaints from the chief complaint string
CC: Abdominal Pain	Derived by keyword matching for “ABDOMINAL PAIN”
CC: Chest Pain	Derived by keyword matching for “CHEST PAIN”
CC: Shortness of Breath	Derived by keyword matching for “SHORTNESS OF BREATH”
CC: Fall	Derived by keyword matching for “FALL”
CC: Fever	Derived by keyword matching for “FEVER”
CC: Critical	Derived by keyword matching for terms including suicide, overdose, trauma, bleeding, seizure, and stroke
Comorbidity Burden (2)	
Charlson Comorbidity Index	Derived using the Quan et al. (2005) ICD-10 mapping algorithm, range 0 to 37
Disease Condition Count	Derived as unique Clinical Classifications Software categories per patient from ICD-10 codes
Home Medications (3)	
Home Medication Count	Derived as count of medication rows per Medical Record Number from the meds.csv table
Unique Medication Classes Count	Derived as distinct Med_class categories per Medical Record Number
Polypharmacy Flag	Derived as binary indicator for home medication count of five or more

144 2.4.1 Hyperparameter Selection

145 Hyperparameters for each model were optimized using Bayesian search with ten fold cross validation
 146 on the training set, targeting the area under the receiver operating characteristic curve metric. The search
 147 space for each model included key parameters such as tree depth, learning rate, and the number of

Table 3. Hyperparameter search spaces and optimal values for Bayesian optimization

Model	Hyperparameter	Search Range	Optimal Value
XGBoost	max depth	3 to 10	6
	learning rate	0.01 to 0.3	0.05
	n estimators	100 to 500	500
	subsample	0.6 to 1.0	0.8
	colsample bytree	-	0.8
	scale pos weight	-	4.26
LightGBM	num leaves	10 to 100	31
	learning rate	0.01 to 0.3	0.05
	n estimators	100 to 500	500
	subsample	0.6 to 1.0	0.8
	colsample bytree	0.6 to 1.0	0.8
	reg alpha	-	0.01
	reg lambda	-	0.01
Random Forest	n estimators	100 to 500	500
	max depth	5 to 20	15
	min samples split	2 to 20	20
	min samples leaf	-	10
	max features	sqrt	sqrt
TabNet	n steps	1 to 5	5
	n d	8 to 64	64
	n a	8 to 64	64
	gamma	1.0 to 2.0	1.5
	lambda sparse	-	1e-4

estimators, as detailed in Table 3. The best hyperparameter configuration was selected based on the mean validation area under the receiver operating characteristic curve averaged across the ten folds. Final model performance was then evaluated on the held out test set.

2.4.2 Cross Validation Procedure

Ten fold cross validation was performed on the training set of 81,588 patients to ensure reliable performance estimates and prevent overfitting. The training data were partitioned into ten approximately equal sized folds while preserving patient level separation, meaning no patient appeared in more than one fold. For each fold, the model was trained on nine folds and validated on the remaining fold. This process was repeated ten times, with each fold serving as the validation set once. The reported cross validation metrics, including accuracy, precision, recall, specificity, F1 score, and area under the receiver operating characteristic curve, represent the mean values across the ten folds along with their standard deviations. After hyperparameter optimization, each model was retrained on the full training set using the optimal hyperparameters and evaluated on the independent test set.

2.4.3 Evaluation Metrics

The quality of the produced models was assessed using six standard classification metrics. Let TP represent true positives, TN represent true negatives, FP represent false positives, and FN represent false negatives. The metrics were defined as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (4)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(FPR) d(FPR) \quad (6)$$

where TPR is the true positive rate (sensitivity) and FPR is the false positive rate (1 - specificity). The area under the receiver operating characteristic curve evaluates the model's capacity to distinguish between positive and negative classes across all classification thresholds.

All analyses were conducted using Python version 3.13 with the scikit learn, XGBoost, LightGBM, and PyTorch libraries.

2.5 Interpretability: SHAP Analysis

SHAP values, which stand for SHapley Additive exPlanations, were computed to identify the most important features for each prediction task [25, 26]. SHAP analysis was conducted using the best performing model for each task. XGBoost was used for emergency department length of stay prediction and hospital admission prediction, while Random Forest was used for hospital length of stay prediction among admitted patients.

2.6 Sensitivity Analyses

To assess model robustness across patient subgroups, we conducted stratified sensitivity analyses. The test set was partitioned by three clinically relevant stratifiers: chief complaint (derived from the seven CC variables in Table 2), age group (<65, 65–74, 75–84, ≥85 years), and triage acuity level (Acuity 1 through 5, with Acuity 4 and 5 combined due to small sample sizes). For each subgroup, we calculated AUC-ROC with 95% confidence intervals using 1,000 bootstrap resamples. Subgroups with fewer than 30 observations were noted as underpowered. These analyses were performed separately for each of the three prediction tasks using the same best-performing models identified in Section 3.1 (XGBoost for ED-LOS and hospital admission, Random Forest for hospital LOS).

3 RESULTS

3.1 Task Performance

Table 4 presents the evaluation metrics for all four ML algorithms across the three prediction tasks. The summarised results show that XGBoost consistently achieved the best performance for ED-LOS and hospital admission prediction, while Random Forest performed best for hospital LOS prediction among admitted patients.

For ED-LOS ≥8 hours, XGBoost achieved the highest accuracy (65.2%) and AUC-ROC (0.660). For hospital admission, XGBoost achieved the highest accuracy (75.6%) and AUC-ROC (0.817). For hospital LOS >7 days among admitted patients, Random Forest achieved the highest accuracy (63.5%) and AUC-ROC (0.671).

Table 4. Evaluation metrics for the four ML algorithms on the test set

Task	Algorithm	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score	AUC-ROC
ED-LOS $\geq 8h$	XGBoost	65.2	68.4	79.1	44.0	0.423	0.660
	LightGBM	62.8	65.2	82.8	38.2	0.416	0.648
	RF	65.0	68.2	71.6	51.9	0.422	0.660
	TabNet	64.5	67.5	66.8	54.6	0.411	0.646
Hospital Admission	XGBoost	75.6	72.7	79.3	68.1	0.696	0.816
	LightGBM	75.4	72.5	79.3	68.0	0.696	0.816
	RF	75.0	72.1	73.5	72.7	0.684	0.812
	TabNet	75.3	72.3	81.3	65.3	0.695	0.811
Hosp-LOS $>7d^*$	XGBoost	62.1	65.0	69.8	53.3	0.387	0.655
	LightGBM	60.2	62.8	58.0	63.4	0.377	0.647
	RF	63.5	66.8	54.8	69.9	0.395	0.670
	TabNet	61.8	64.5	54.3	68.4	0.383	0.651

*Among admitted patients only (test set n=4,765)

3.2 Subgroup Sensitivity Analyses

Table 5 presents model performance stratified by chief complaint, age group, and triage acuity level. Hospital admission prediction remained stable across all subgroups, with AUC values ranging from 0.704 (patients aged ≥ 85 years) to 0.862 (fever complaints). In contrast, ED-LOS prediction showed declining discrimination with increasing age, from 0.688 in patients under 65 years to 0.582 in patients aged ≥ 85 years. Performance for prolonged hospital LOS among admitted patients was broadly consistent across subgroups, though small sample sizes in the highest acuity category (Acuity 1, n=33) warrant cautious interpretation.

3.3 SHAP Analysis

Figure 2 presents the SHAP summary plot for XGBoost for ED-LOS $\geq 8h$ prediction (Task 1). The top five features by mean absolute SHAP value were: EMS arrival, age, triage acuity level, disease condition count and home medication count. Higher EMS arrival values (red) increased predicted risk of prolonged clinic stay, consistent with EMS-transported patients having higher acuity.

Figure 3 presents the SHAP summary plot for XGBoost for hospital admission prediction (Task 2). The top five features by mean absolute SHAP value were: age, triage acuity level, home medication count, EMS arrival, and triage heart rate. Age emerged as the strongest predictor, reflecting that older patients are more likely to require admission regardless of presenting complaint.

Figure 4 presents the SHAP summary plot for Random Forest for hosp-LOS $>7d$ prediction among admitted patients (Task 3). The top five features by mean absolute SHAP value were: triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count. Vital signs played a larger role here, indicating that physiological instability at triage predicts prolonged hospitalization.

Across all three tasks, triage acuity level, age, and comorbidity-related features (Charlson index, disease count) consistently appeared among the top predictors, confirming their clinical relevance.

Beyond these common predictors, each task had unique top features that reflect different stages of patient care. For ED-LOS prediction, EMS arrival emerged as a key predictor, suggesting that patients requiring ambulance transport face longer clinic stay times due to higher acuity and complex care coordination. For hospital admission prediction, triage heart rate was uniquely important, indicating that physiological stress at presentation strongly influences disposition decisions. For prolonged hospitalization prediction among admitted patients, triage systolic blood pressure and triage heart rate dominated, showing that vital sign abnormalities at triage signal complex inpatient courses requiring extended hospital stays. These differences suggest that while baseline demographic and comorbidity factors are universally important, physiological and logistical factors become more relevant at specific decision points.

Table 5. Subgroup sensitivity analysis results across all three prediction tasks

Subgroup	AUC-ROC (95% CI)		
	ED-LOS ≥ 8 h	Hospital Admission	Hosp-LOS > 7 d
Chief Complaint			
Abdominal Pain	0.622 (n=1,619)	0.762 (n=1,619)	0.682 (n=581)
Chest Pain	0.679 (n=926)	0.807 (n=926)	0.661 (n=394)
Critical	0.657 (n=1,377)	0.804 (n=1,377)	0.667 (n=524)
Fall	0.606 (n=688)	0.779 (n=688)	0.664 (n=386)
Fever	0.643 (n=410)	0.862 (n=410)	0.629 (n=162)
Other/Unspecified	0.675 (n=7,466)	0.821 (n=7,466)	0.677 (n=2,883)
Shortness of Breath	0.615 (n=658)	0.824 (n=658)	0.686 (n=298)
Age Group			
<65 years	0.688 (n=8,058)	0.805 (n=8,058)	0.696 (n=2,336)
65–74 years	0.613 (n=1,739)	0.762 (n=1,739)	0.664 (n=948)
75–84 years	0.598 (n=1,374)	0.718 (n=1,374)	0.669 (n=902)
≥ 85 years	0.582 (n=849)	0.704 (n=849)	0.613 (n=579)
Triage Acuity			
Acuity 1 (Resuscitation)	0.541 (n=95)	0.770 (n=95)	0.693 (n=33)*
Acuity 2 (Emergent)	0.610 (n=3,180)	0.774 (n=3,180)	0.645 (n=1,347)
Acuity 3 (Urgent)	0.650 (n=7,510)	0.757 (n=7,510)	0.684 (n=3,019)
Acuity 4–5 (Semi/Non-urgent)	0.714 (n=1,235)	0.729 (n=1,235)	0.719 (n=366)
Overall Performance			
Overall	0.667	0.816	0.675

Note: Values shown are AUC-ROC with sample sizes in parentheses.

4 DISCUSSION

ED-LOS represents a crucial key indicator of the efficiency of healthcare services. The ability to understand the reasons for prolonged ED-LOS could support the detection of bottlenecks in ED organisation. This study demonstrates that machine learning using only triage-available data can accurately predict three critical ED outcomes.

Our results indicate that patients with prolonged ED-LOS belong mainly to the over 64 years age group. Moreover, patients with prolonged hospital LOS also tended to be older and have higher comorbidity burden. These findings align with previous studies [14, 16, 17].

The evaluation metrics showed that XGBoost and Random Forest consistently outperformed the other algorithms. For ED-LOS prediction, XGBoost achieved the highest accuracy of 65.2% and AUC-ROC of 0.660. For hospital admission prediction, XGBoost achieved the highest accuracy of 75.6% and AUC-ROC of 0.817. For hospital LOS prediction among admitted patients, Random Forest achieved the highest accuracy of 63.5% and AUC-ROC of 0.671.

Comparing with other ML-based studies, Ricciardi et al. [14] achieved accuracy of 74.8% for predicting ED-LOS greater than 3 hours using Random Forest. Our study achieved comparable performance (65.2% accuracy) for a more stringent threshold of 8 hours. Naemi et al. [4] achieved accuracy ranging from 75% to 82% for ED-LOS prediction. Our results are consistent with these findings.

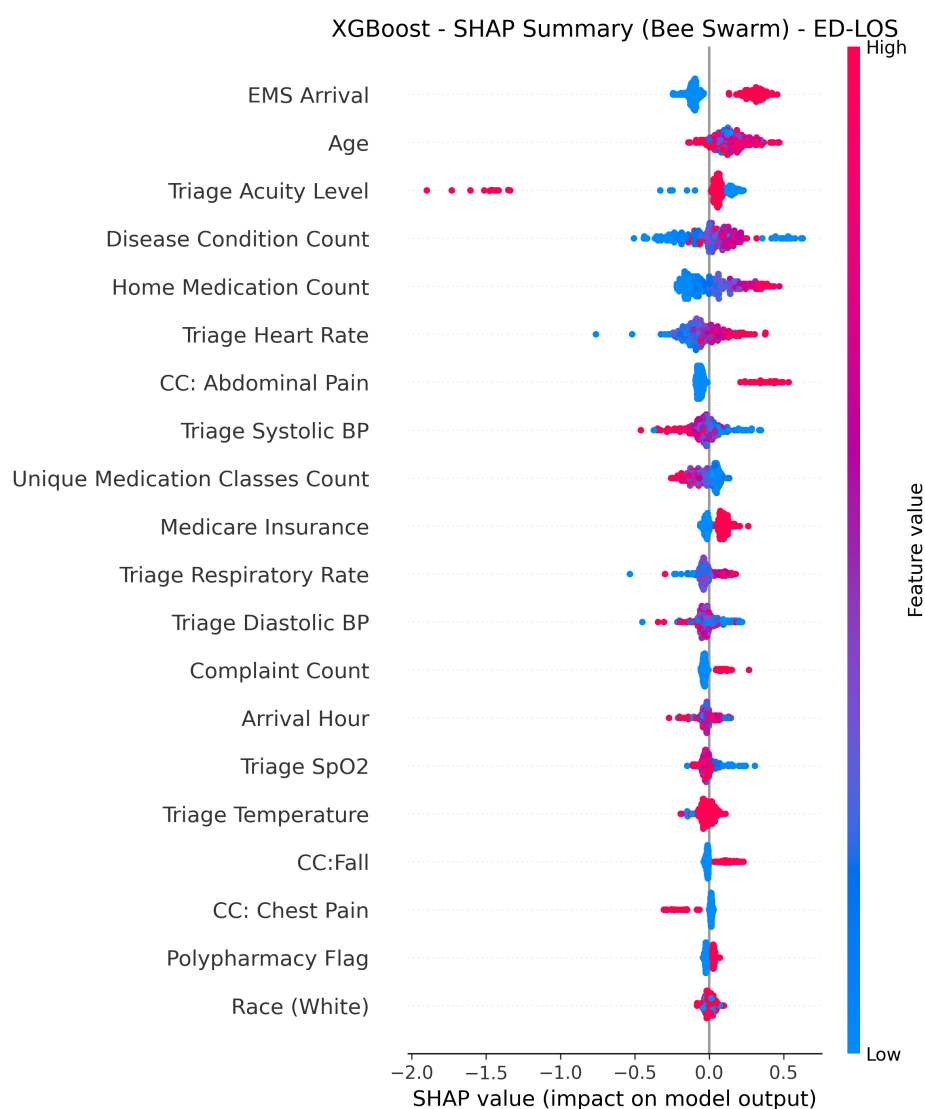


Figure 2. SHAP summary plot for XGBoost for ED-LOS ≥ 8 h prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). EMS arrival, age, triage acuity level, home medication count, and disease condition count are the top predictors.

242 A key contribution of this study is SHAP analysis for interpretability [26, 29]. Triage acuity level,
 243 age, and comorbidity burden were consistently important across all three tasks. Beyond these common
 244 predictors, task-specific features emerged. EMS arrival uniquely predicted prolonged ED clinic stay,
 245 reflecting that ambulance-transported patients require more resources before disposition. Triage heart rate
 246 strongly influenced admission decisions, indicating that physiological stress drives disposition choices.
 247 Vital signs (systolic blood pressure and heart rate) dominated prolonged hospitalization predictions,
 248 suggesting that physiological instability at triage signals complex inpatient courses. These findings
 249 demonstrate how different clinical factors become relevant at different stages of patient care.

250 Subgroup sensitivity analyses revealed stable hospital admission performance across chief complaints
 251 (AUC range: 0.704–0.862) and triage acuity levels, though ED-LOS prediction declined with age (AUC
 252 0.688 in patients < 65 years to 0.582 in patients ≥ 85 years). Missing data were handled using Multiple
 253 Imputation by Chained Equations with 10 imputations, excluding outcome variables to prevent leakage.

254 The results of this study can be exploited to develop a preventive plan to optimise ED management. By
 255 predicting prolonged ED-LOS, decision makers could allocate additional medical and nursing staff during

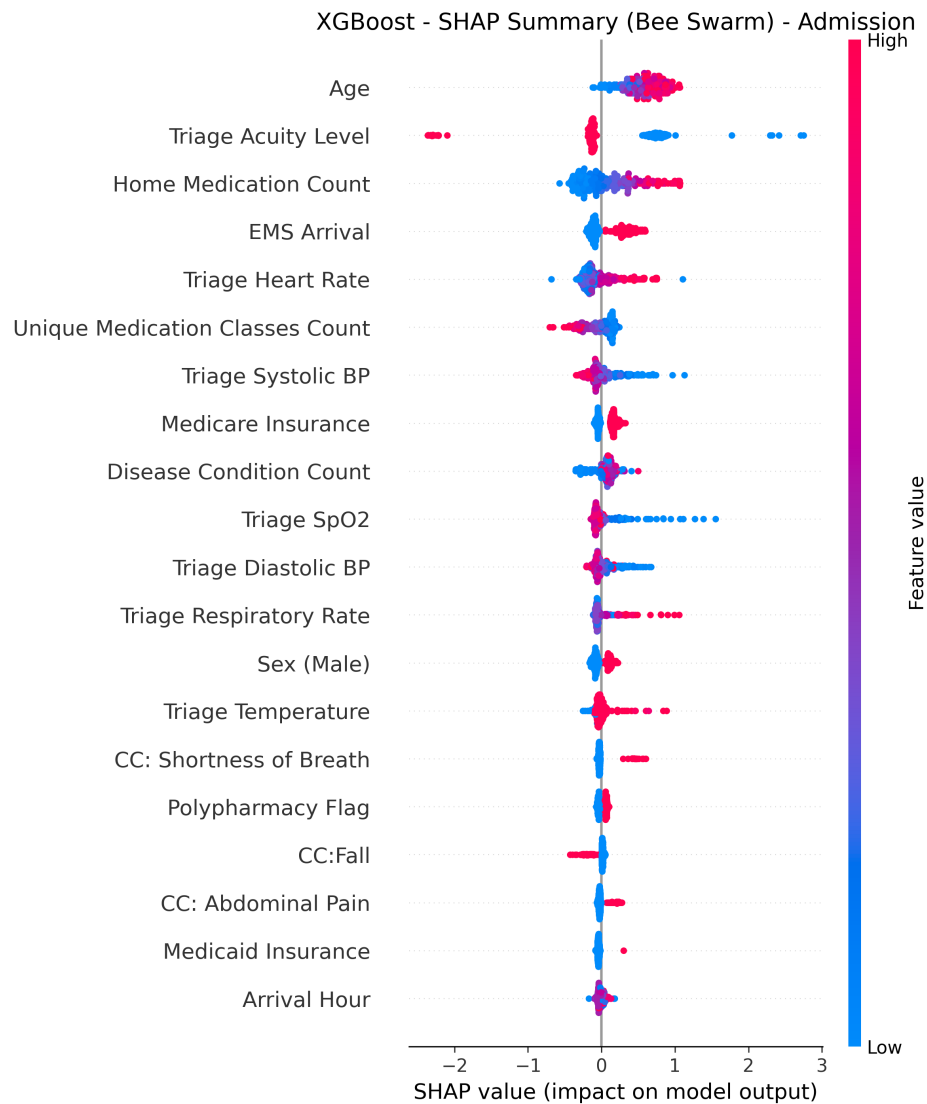


Figure 3. SHAP summary plot for XGBoost for hospital admission prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Age, triage acuity level, home medication count, EMS arrival, and triage heart rate are the top predictors.

critical time slots. By predicting hospital admission, bed management could be improved. By predicting prolonged hospital LOS, discharge planning could be initiated earlier.

4.1 Limitations

This study has limitations that should be acknowledged. First, it is a single-centre study using data from one academic medical centre in California, which limits generalizability to community hospitals, rural settings, and international healthcare systems [54, 56]. Variables such as Medicaid insurance are US-specific, and chief complaint categorisation may differ across settings. External validation across diverse healthcare systems is necessary before widespread deployment, and models would likely require recalibration or retraining for new settings.

Second, while our models achieved acceptable discrimination (AUC up to 0.817), the practical utility of 65%–75% accuracy for ED-LOS prediction means these predictions are intended to supplement, not replace, clinical judgment. The retrospective design also cannot capture clinician acceptance, workflow

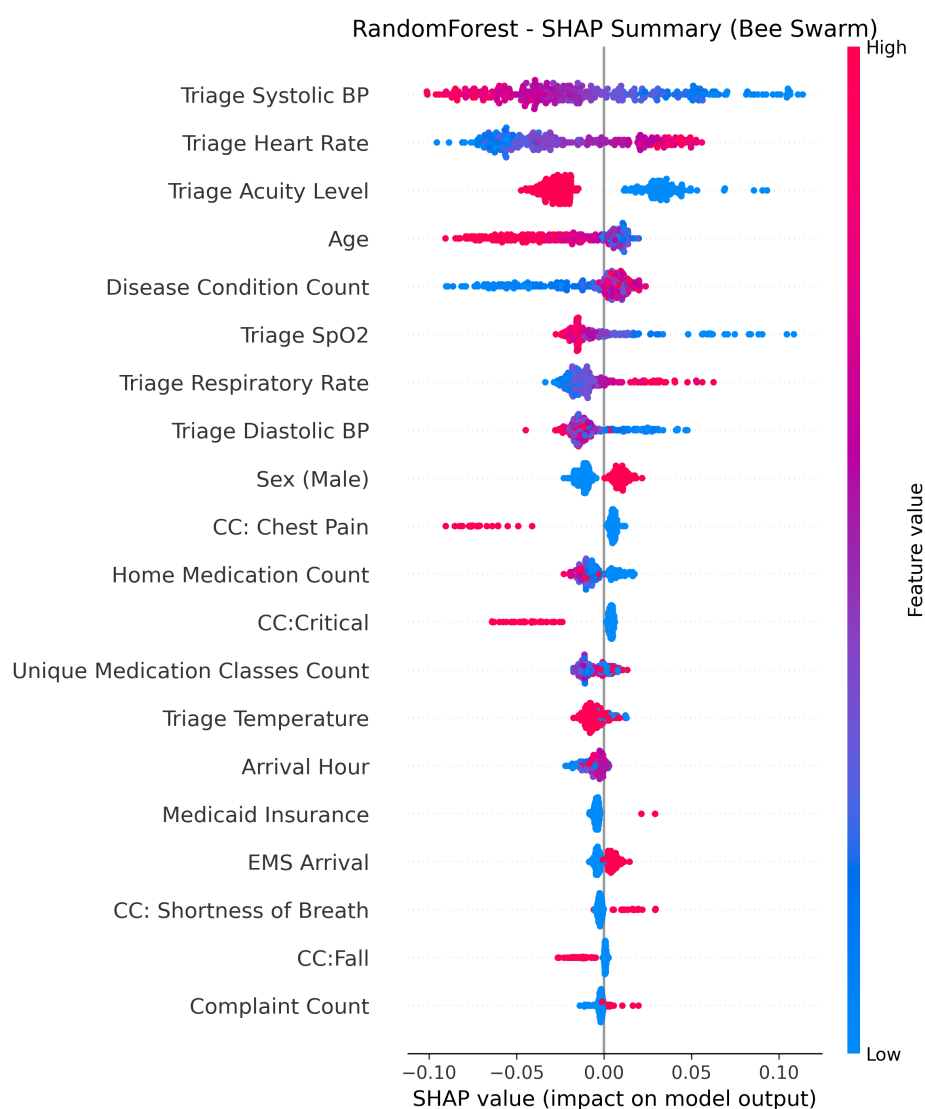


Figure 4. SHAP summary plot for Random Forest for hospital LOS > 7 days prediction among admitted patients. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count are the top predictors.

integration challenges, or alert fatigue [52, 53]. Prospective implementation studies are essential to confirm real-world performance.

Third, the data span from 2020 to 2022, encompassing the COVID-19 pandemic period when ED operations and patient acuity differed substantially from baseline, which may affect temporal validity [10].

5 CONCLUSION

This study demonstrates that machine learning using only triage-available data can accurately predict three critical outcomes in emergency care: prolonged ED clinic stay (≥ 8 hours), hospital admission, and prolonged hospitalization (> 7 days) among admitted patients. Using the MC-MED dataset of 118,385 adult ED visits, XGBoost consistently outperformed other models for ED-LOS and admission prediction, while Random Forest achieved the best performance for hospital LOS prediction.

SHAP analysis revealed that triage acuity level, age, and comorbidity burden were consistently important across all three tasks. Beyond these common predictors, EMS arrival uniquely predicted prolonged ED clinic stay, triage heart rate strongly influenced admission decisions, and vital sign abnormalities dominated predictions of prolonged hospitalization. These findings highlight how different clinical factors become relevant at different stages of patient care.

Real-time deployment of these models at triage could enable early identification of at-risk patients, facilitating proactive bed management and resource allocation. The interpretability of SHAP explanations supports clinician trust and adoption by providing transparent reasons for each prediction. This study provides an evidence base for deploying machine learning based clinical decision support in emergency departments to improve patient flow and optimize resource allocation.

6 CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 AUTHOR CONTRIBUTIONS

Onyedikachi I. Onwurah designed the study, implemented all machine learning models, performed the data analysis, and drafted the manuscript. Sol Richardson contributed to study design, data interpretation, and critical revision. Wu Sheng served as clinical reviewer. Xia Tong supervised the research, guided the methodological approach, reviewed all versions of the manuscript, and approved the final submission.

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9 ETHICS STATEMENT

The Stanford University Institutional Review Board approved the study protocol with a waiver of informed consent. The MC-MED dataset is de-identified and publicly available via PhysioNet.

10 DATA AVAILABILITY STATEMENT

The MC-MED dataset is publicly available on PhysioNet at <https://physionet.org/content/mc-med/1.0.1/> [10]. Analysis code is available at <https://github.com/Drglazizzo/ED-visit-12ML-prediction>.

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12 PUBLISHER'S NOTE

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